EMC First Midterm — Study Notes

Course: Electromagnetic Compatibility, NTUST
Target audience: Student familiar with EM fields and radio-wave basics
Source of truth: chap1.pdf – chap4.pdf

Chapter 1: Introduction to EMC

What Is EMC?

A system is **electromagnetically compatible** when it: 1. Does not cause interference with other systems 2. Is not susceptible to emissions from other systems 3. Does not interfere with itself

(chap1.pdf, p.4)

The Three-Element Model

Every EMC problem reduces to:

$$\text{Source} \rightarrow \text{Coupling Path} \rightarrow \text{Receptor}$$

(chap1.pdf, p.5)

Interference occurs only when the received energy causes the receptor to behave undesirably. The most cost-effective fix attacks the source first, then the coupling path, and last the receptor. *(chap1.pdf, p.6)*

Four EMC Subproblems

#	Problem	Transfer mechanism
1	Radiated emissions	Source radiates outward
2	Radiated susceptibility	External radiation couples in
3	Conducted emissions	Noise exits via power/signal cable
4	Conducted susceptibility	Noise enters via power/signal cable

(chap1.pdf, p.7)

Other EMC Phenomena (Not Government-Regulated)

ESD, EMP, lightning, and TEMPEST all fall outside standard emission limits but must be designed against. *(chap1.pdf, p.9)*

Electrical Dimensions and Wave Propagation

Why it matters: A structure's ability to radiate scales with its electrical size (size relative to wavelength), not its physical size.

Time delay for a signal to traverse length $\mathscr{L}$:

$$T_D = \frac{\mathscr{L}}{v}$$

(chap1.pdf, p.16)

Phase velocity (relates wavelength and frequency):

$$v = \lambda f \quad \text{m/s}$$

(chap1.pdf, p.18)

Velocity in a medium (slows in denser dielectric):

$$v = \frac{1}{\sqrt{\epsilon\mu}} = \frac{v_0}{\sqrt{\epsilon_r \mu_r}}, \quad v_0 = 3 \times 10^8 \text{ m/s}$$

(chap1.pdf, p.20)

Electrical dimension (dimensionless measure of electrical size):

$$k = \frac{\mathscr{L}}{\lambda}$$

(chap1.pdf, p.21)

Rule of thumb: if $k < 0.1$ ($\mathscr{L} < \lambda/10$), lumped-circuit analysis is adequate. If $k \sim 1$ or larger, transmission-line or full-wave analysis is needed.

Key frequencies and wavelengths in free space *(chap1.pdf, p.21)*:

Frequency	λ
30 MHz	10 m
300 MHz	1 m
3 GHz	10 cm

PCB substrate: Epoxy glass ($\epsilon_r = 4.7$), so $\lambda_{\text{PCB}} = \lambda_0/\sqrt{4.7} \approx \lambda_0/2.17$. *(chap1.pdf, p.23)*

Decibels

Power gain:

$$G_{\mathrm{dB}} = 10\log_{10}\left(\frac{P_{\mathrm{out}}}{P_{\mathrm{in}}}\right)$$

(chap1.pdf, p.25)

Voltage/current gain:

$$G_{\mathrm{dB}} = 20\log_{10}\left(\frac{V_{\mathrm{out}}}{V_{\mathrm{in}}}\right) = 20\log_{10}\left(\frac{I_{\mathrm{out}}}{I_{\mathrm{in}}}\right)$$

(chap1.pdf, p.26)

Handy values: ratio of $2 \to 6$ dB (voltage), 3 dB (power); ratio of $10 \to 20$ dB (voltage), 10 dB (power). *(chap1.pdf, p.27)*

Gains add in dB — useful for cascaded amplifiers/attenuators. *(chap1.pdf, p.28)*

Cable loss formula (for matched load, $Z_L = Z_C$):

$$\text{Cable loss (dB)} = 8.686\,\alpha\mathscr{L}$$

(chap1.pdf, p.30)

where α (Np/m) is the attenuation constant. This formula appears again in Chapter 4.

Unit conversion (for 50 Ω system):

$$\text{dB}\mu\text{V (RMS)} = 107 + \text{dBm}$$

(chap1.pdf, p.30)

Chapter 2: EMC Requirements

Regulatory Overview

Region	Body	Standard
USA	FCC	Part 15 of Title 47 CFR
International	CISPR (IEC committee)	CISPR 22
Military	DoD	MIL-STD-461E

(chap2.pdf, p.4–5)

Class A = commercial/industrial (less stringent).
Class B = residential (more stringent, because residents lack technical resources).
(chap2.pdf, p.6)

Conducted Emissions (150 kHz – 30 MHz)

Measured via **LISN** (Line Impedance Stabilization Network). The LISN presents ~50 Ω to the EUT and blocks 60 Hz power from reaching the receiver. *(chap2.pdf, p.28–30)*

Key Class B limits: 56–60 dBμV (quasi-peak).
Key Class A limits: 73–79 dBμV (quasi-peak). *(chap2.pdf, p.8)*

CISPR 22 conducted limits are the same as FCC. *(chap2.pdf, p.15)*

Radiated Emissions (30 MHz – 40 GHz)

Measured in a **Semianechoic Chamber (SAC)** or **OATS**.
Both vertical and horizontal polarizations measured; antenna scanned 1–4 m height. *(chap2.pdf, p.26)*

Class B at **3 m**: 40 dBμV/m (30–88 MHz) to 54 dBμV/m (>960 MHz). *(chap2.pdf, p.11)*
Class A at **10 m**: 39 dBμV/m (30–88 MHz) to 49.5 dBμV/m (>960 MHz). *(chap2.pdf, p.11)*

Distance normalization: To compare limits measured at different distances:

$$\Delta E_{\mathrm{dB}} = 20 \log_{10}\left(\frac{d_1}{d_2}\right)$$

(chap2.pdf, p.13)

Normalizing Class A (10 m) to 3 m adds $20 \log_{10}(10/3) \approx 10.5$ dB — confirming Class A is about 10 dB less stringent than Class B. *(chap2.pdf, p.13)*

CISPR 22 radiated limits cover only up to **1 GHz** (FCC goes to 40 GHz). European CISPR 22 Class B limits (30 dBμV/m at 10 m, 30–230 MHz) are more stringent than FCC Class B. *(chap2.pdf, p.14)*

MIL-STD-461E

Much wider frequency ranges and can be **tailored** (waived) by procuring authority. Key test families:

- **CE** = Conducted Emissions; **CS** = Conducted Susceptibility
- **RE** = Radiated Emissions; **RS** = Radiated Susceptibility

CE102 covers 10 kHz–10 MHz on power leads. RE102 covers 10 kHz–18 GHz. RS103 sets up to 200 V/m for external aircraft. *(chap2.pdf, p.17–22)*

Design Strategy

- Find the **dominant emission source** before applying fixes. *(chap2.pdf, p.40)*
- Plan for EMC early: "Paper work before layout." Provide passive component footprints (capacitors, resistors) as an insurance policy. *(chap2.pdf, p.41)*
- Even a simple PCB with a 10 MHz oscillator can exceed Class B radiated limits by **30 dB**. *(chap2.pdf, p.38)*

Chapter 3: Signal Spectra

Why the Spectrum Matters for EMC

The **rise time** (τ_r), not the clock frequency, dominates how much high-frequency energy a signal produces. Faster edges → broader spectrum → harder to pass EMC. *(chap3.pdf, p.3)*

Fourier Series

Any periodic signal with period T and fundamental frequency $f_0 = 1/T$:

$$x(t) = \sum_{n=-\infty}^{\infty} c_n \, e^{jn\omega_0 t}$$

(chap3.pdf, p.7)

Fourier coefficients:

$$c_n = \frac{1}{T} \int_{t_1}^{t_1+T} x(t) \, e^{-jn\omega_0 t} \, dt$$

(chap3.pdf, p.8)

c_0 is the DC (average) value; $c_n = c_{-n}^*$ for real signals. The one-sided spectrum uses $2|c_n|$ for $n \geq 1$. *(chap3.pdf, p.9)*

Spectrum of a Rectangular Pulse Train

For a train of rectangular pulses, amplitude A, width τ, period T:

$$|c_n| = \frac{A\tau}{T} \left| \frac{\sin(\pi f \tau)}{\pi f \tau} \right|$$

(chap3.pdf, p.12)

DC level: $c_0 = A\tau/T = AD$ where $D = \tau/T$ is duty cycle. *(chap3.pdf, p.12)*

50% duty cycle: Even harmonics are zero ($c_n = 0$ for even n). *(chap3.pdf, p.14)*

Spectrum of a Trapezoidal (Clock) Waveform

A trapezoidal pulse has finite rise time τ_r and fall time τ_f. For $\tau_r = \tau_f$, the coefficients are the **product of two sinc functions**:

$$c_n = A\frac{\tau}{T} \cdot \frac{\sin\left(\frac{1}{2}n\omega_0\tau\right)}{\frac{1}{2}n\omega_0\tau} \cdot \frac{\sin\left(\frac{1}{2}n\omega_0\tau_r\right)}{\frac{1}{2}n\omega_0\tau_r} \cdot e^{-jn\omega_0(\tau+\tau_r)/2}$$

(chap3.pdf, p.29)

Spectral Envelope (Bode Plot Approximation)

The magnitude spectrum envelope has **three regions**:

Frequency range	Slope	Breakpoint
$f < f_1 = \dfrac{1}{\pi\tau}$	0 dB/decade (flat)	f_1
$f_1 < f < f_2 = \dfrac{1}{\pi\tau_r}$	-20 dB/decade	f_2
$f > f_2$	-40 dB/decade	—

(chap3.pdf, p.34)

Peak envelope level: $2A\tau/T$. The first breakpoint f_1 depends on pulse width τ; the second breakpoint f_2 depends on rise time τ_r.

> **Key insight:** To reduce high-frequency emissions, increase the rise/fall time τ_r. This moves f_2 lower and steepens the roll-off sooner. *(chap3.pdf, p.35)*

Signal Bandwidth

$$\text{BW} = \frac{1}{\tau_r}$$

(chap3.pdf, p.37)

This is the frequency at which the high-frequency (-40 dB/dec) region begins; above this frequency the spectrum falls very fast.

Effect of Duty Cycle

Reducing duty cycle $D = \tau/T$: - Lowers the flat (DC) spectral level - Moves the first breakpoint $f_1 = 1/(\pi\tau)$ to higher frequency - Does **not** change the high-frequency content (no effect on f_2 or the -40 dB/dec slope)

(chap3.pdf, p.38–39)

Ringing (Undershoot/Overshoot)

PCB trace inductance and capacitance cause ringing oscillations at frequency ω_r on signal edges. In the spectrum, ringing adds a **spectral peak at** ω_r. Fix with a series damping resistor or ferrite bead. *(chap3.pdf, p.40–41)*

Spectrum Analyzers

A spectrum analyzer is a swept bandpass receiver. The **resolution bandwidth** must be smaller than the harmonic spacing to resolve individual harmonics. Regulatory measurements use a **quasi-peak (QP) detector** (penalizes pulsed interference more than CW). *(chap3.pdf, p.44–46)*

PCM-NRZ Power Spectral Density

For random binary data (0 to X_0, equal probability):

$$G_x(f) = \frac{X_0^2}{4}\,\delta(f) + \frac{X_0^2 T}{4}\,\frac{\sin^2(\pi f T)}{(\pi f T)^2} \quad \text{W/Hz}$$

(chap3.pdf, p.52)

The first term is a DC spike; the second is a continuous sinc-squared spectrum with nulls at $f = n/T$.

Chapter 4: Transmission Lines and Signal Integrity

Why Transmission Lines?

At high frequencies, a PCB trace is not a simple wire — it is a **transmission line** that: - Imposes a **time delay** $T_D = \mathscr{L}/v$ *(chap4.pdf, p.3)* - Can cause **reflections** when impedance is mismatched $\rightarrow$ signal integrity degradation

Telegrapher's Equations

A lossless transmission line is modeled as cascaded $l\Delta z$ inductors (series) and $c\Delta z$ capacitors (shunt), where l (H/m) and c (F/m) are per-unit-length parameters.

KVL $\rightarrow$ voltage equation:

$$\frac{\partial V}{\partial z} = -l\,\frac{\partial I}{\partial t}$$

(chap4.pdf, p.8)

KCL → current equation:

$$\frac{\partial I}{\partial z} = -c\,\frac{\partial V}{\partial t}$$

(chap4.pdf, p.9)

Combined into wave equations:

$$\frac{\partial^2 V}{\partial z^2} = lc\,\frac{\partial^2 V}{\partial t^2}, \quad \frac{\partial^2 I}{\partial z^2} = lc\,\frac{\partial^2 I}{\partial t^2}$$

(chap4.pdf, p.10)

Key Relationships

For a **homogeneous** medium:

$$lc = \mu\epsilon, \quad v = \frac{1}{\sqrt{lc}} = \frac{v_0}{\sqrt{\epsilon_r}}, \quad Z_C = \sqrt{\frac{l}{c}}$$

(chap4.pdf, p.12, 28)

TEM waves are supported: no field components along the direction of propagation.

Time-Domain Solution

General solution (forward + backward waves):

$$V(z,t) = V^+\Big(t - \frac{z}{v}\Big) + V^-\Big(t + \frac{z}{v}\Big)$$

$$I(z,t) = \frac{1}{Z_C}V^+\Big(t - \frac{z}{v}\Big) - \frac{1}{Z_C}V^-\Big(t + \frac{z}{v}\Big)$$

(chap4.pdf, p.34)

Initially launched voltage from source (voltage divider between R_S and Z_C):

$$V(0,t) = \frac{Z_C}{R_S + Z_C}\,V_S(t)$$

(chap4.pdf, p.37)

Reflection Coefficients

Load reflection coefficient:

$$\Gamma_L = \frac{R_L - Z_C}{R_L + Z_C}$$

(chap4.pdf, p.35)

Source reflection coefficient:

$$\Gamma_S = \frac{R_S - Z_C}{R_S + Z_C}$$

(chap4.pdf, p.37)

Special cases:

Load	Γ_L
Matched ($R_L = Z_C$)	0 — no reflection
Open circuit ($R_L \to \infty$)	+1 — full reflection
Short circuit ($R_L = 0$)	-1 — full reflection, reversed

Bounce (Lattice) Diagram

A graphical tool: plot z on the horizontal axis, time on vertical, and trace diagonal lines for each forward/backward wave. Multiply voltage by Γ_L at the load end and Γ_S at the source end with each bounce. *(chap4.pdf, p.42–43)*

Steady-state voltage on a lossless line equals the DC voltage divider result $V_S \cdot R_L/(R_S + R_L)$ regardless of Z_C — after enough reflections, the voltage settles.

When Can the Line Be Ignored?

A transmission line can be treated as a lumped element (just a wire) when:

$$\tau_r > 10\,T_D$$

(chap4.pdf, p.59)

where τ_r is the signal rise time and $T_D = \mathscr{L}/v$ is the one-way travel time. If $\tau_r < 10\,T_D$, transmission-line effects (reflections, overshoot) are significant.

Matching Schemes

Series match: Insert resistor R at the source so that $R_S + R = Z_C$. *(chap4.pdf, p.54)*
- Eliminates ringing; zero steady-state power dissipation in R.

Parallel match: Add resistor R at load so that $R \| R_L = Z_C$. *(chap4.pdf, p.55)*
- Eliminates ringing; dissipates power continuously.

Capacitive and Inductive Terminations

A **capacitive load** looks initially like a short ($\Gamma_L = -1$) then transitions to open ($\Gamma_L = +1$). Added delay at 50% threshold:

$$t_d = 0.693\, Z_C\, C$$

(chap4.pdf, p.52)

An **inductive load** is initially open ($\Gamma_L = +1$), then short-circuits ($\Gamma_L = -1$). *(chap4.pdf, p.53)*

Phasor (Sinusoidal Steady-State) Analysis

For sinusoidal excitation at frequency f, with phase constant $\beta = \omega/v = 2\pi/\lambda$:

$$\hat{V}(z) = \hat{V}^+ e^{-j\beta z} + \hat{V}^- e^{j\beta z}$$

(chap4.pdf, p.73)

Input impedance of a lossless line of length $\mathscr{L}$ with load $\hat{Z}_L$:

$$\hat{Z}_{\text{in}} = Z_C \frac{\hat{Z}_L + jZ_C \tan \beta\mathscr{L}}{Z_C + j\hat{Z}_L \tan \beta\mathscr{L}}$$

(chap4.pdf, p.76)

VSWR (measures the severity of mismatch):

$$\text{VSWR} = \frac{1 + |\hat{\Gamma}_L|}{1 - |\hat{\Gamma}_L|}, \quad 1 \leq \text{VSWR} \leq \infty$$

(chap4.pdf, p.77)

VSWR = 1 means perfectly matched; VSWR = ∞ means total reflection (open or short load).

Average power flow:

$$P_{\text{av}} = \frac{|\hat{V}^+|^2}{2Z_C}\left(1 - |\hat{\Gamma}_L|^2\right)$$

(chap4.pdf, p.78)

Lossy Transmission Lines

Add per-unit-length resistance r (skin effect) and conductance g (dielectric loss):

Propagation constant:

$$\hat{\gamma} = \sqrt{(r + j\omega l)(g + j\omega c)} = \alpha + j\beta$$

(chap4.pdf, p.81)

Skin depth (current penetration depth in conductor):

$$\delta = \frac{1}{\sqrt{\pi f \mu_0 \sigma}} \text{ m}$$

(chap4.pdf, p.82)

Skin depth decreases as $1/\sqrt{f}$, so conductor resistance increases as $\sqrt{f}$ at high frequency (10 dB/decade). *(chap4.pdf, p.82–83)*

Loss tangent (dielectric loss):

$$\tan\theta = \frac{\epsilon_r''}{\epsilon_r'}, \quad g = \omega c \tan\theta$$

(chap4.pdf, p.87)

Dielectric conductance g increases linearly with frequency (20 dB/decade). For FR-4: $\tan\theta \approx 0.02$.

Low-loss approximation (valid when $r \ll \omega l$ and $g \ll \omega c$):

$$\alpha \approx \frac{1}{2}\left(\frac{r}{Z_C} + gZ_C\right), \quad \beta \approx \omega\sqrt{lc}, \quad Z_C \approx \sqrt{\frac{l}{c}}$$

(chap4.pdf, p.90–91)

In this regime β is linear in f (no phase dispersion) and Z_C is real and frequency-independent.

Cable loss (matched load):

$$\text{Loss (dB)} = 8.686\,\alpha\mathscr{L}$$

(chap4.pdf, p.89)

Quick-Reference Formula Sheet

Concept	Formula	Source
Time delay	$T_D = \mathscr{L}/v$	*(chap1.pdf, p.16)*
Phase velocity	$v = \lambda f = v_0/\sqrt{\epsilon_r \mu_r}$	*(chap1.pdf, p.18, 20)*
Electrical dimension	$k = \mathscr{L}/\lambda$	*(chap1.pdf, p.21)*
Voltage gain (dB)	$20\log_{10}(V_{\text{out}}/V_{\text{in}})$	*(chap1.pdf, p.26)*
Cable loss	$8.686\alpha\mathscr{L}$ dB	*(chap1.pdf, p.30)*
Distance normalization	$\Delta E = 20\log_{10}(d_1/d_2)$ dB	*(chap2.pdf, p.13)*
Fourier coefficient	$c_n = \frac{1}{T}\int x(t)e^{-jn\omega_0 t}dt$	*(chap3.pdf, p.8)*
Rect. pulse spectrum	$\lVert c_n \rVert = \frac{A\tau}{T}\left\lVert \frac{\sin \pi f\tau}{\pi f\tau} \right\rVert$	*(chap3.pdf, p.12)*
First breakpoint	$f_1 = 1/(\pi\tau)$	*(chap3.pdf, p.34)*
Second breakpoint	$f_2 = 1/(\pi\tau_r)$	*(chap3.pdf, p.34)*
Signal bandwidth	$\text{BW} = 1/\tau_r$	*(chap3.pdf, p.37)*
PCM-NRZ PSD	$G_x(f) = \frac{X_0^2}{4}\delta(f) + \frac{X_0^2 T}{4}\frac{\sin^2 \pi f T}{(\pi f T)^2}$	*(chap3.pdf, p.52)*
Telegrapher's (KVL)	$\partial V/\partial z = -l\,\partial I/\partial t$	*(chap4.pdf, p.8)*
Telegrapher's (KCL)	$\partial I/\partial z = -c\,\partial V/\partial t$	*(chap4.pdf, p.9)*
Characteristic impedance	$Z_C = \sqrt{l/c}$	*(chap4.pdf, p.28)*
lc product	$lc = \mu\epsilon$, so $v = 1/\sqrt{lc}$	*(chap4.pdf, p.12)*
Load reflection	$\Gamma_L = (R_L - Z_C)/(R_L + Z_C)$	*(chap4.pdf, p.35)*
Source reflection	$\Gamma_S = (R_S - Z_C)/(R_S + Z_C)$	*(chap4.pdf, p.37)*
Launched voltage	$V(0,t) = \frac{Z_C}{R_S + Z_C}V_S(t)$	*(chap4.pdf, p.37)*
TL negligible	$\tau_r > 10T_D$	*(chap4.pdf, p.59)*
Input impedance	$Z_{\text{in}} = Z_C\frac{Z_L + jZ_C\tan\beta\mathscr{L}}{Z_C + jZ_L\tan\beta\mathscr{L}}$	*(chap4.pdf, p.76)*
VSWR	$(1 + \lVert\Gamma_L\rVert)/(1 - \lVert\Gamma_L\rVert)$	*(chap4.pdf, p.77)*
Propagation constant	$\hat{\gamma} = \sqrt{(r + j\omega l)(g + j\omega c)}$	*(chap4.pdf, p.81)*
Skin depth	$\delta = 1/\sqrt{\pi f \mu_0 \sigma}$	*(chap4.pdf, p.82)*
Loss tangent	$\tan\theta = \epsilon_r''/\epsilon_r'$, $g = \omega c\tan\theta$	*(chap4.pdf, p.87)*
Low-loss α	$\alpha \approx \frac{1}{2}(r/Z_C + gZ_C)$	*(chap4.pdf, p.91)*